GLOBAL METHANE EMISSIONS ANALYSIS REPORT (2022)

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CHAPTER ONE

1.0 INTRUDUCTION

1.1 BACKGROUND

Climate change is a global concern that needs not to be neglected because of its impact on human existence. It is a long-term shift in temperature and weather pattern. Though it can be a natural phenomenon due to volcanic eruption and sun's activities, adverse human activities like burning of fossil fuel have greatly contributed to the change in temperatures and weather conditions.

The burning of fossil fuels like coal, crude oil and natural gas releases gases, such as CO2 and CH4, which act like blankets that trap heat near the earth surface and prevent it from being released into the atmosphere; these gases are known as greenhouse gases.

The existence of greenhouse gases is of great importance if kept in the right proportion. The earth is meant to release all its absorbed heat back into the atmosphere, therefore, causing it to go below -18°C, which is very cold for human habitation. These greenhouse gases help to trap some proportion of this heat to keep the earth surface warm. Due to the daily activities of burning and largely emitting these gases in great proportions, the enormous amounts of heat absorbed are trapped back and are not allowed to be released, therefore, causing global warming.

Methane (CH4) is a colorless, odorless, highly flammable gas made of one carbon atom and four hydrogen atoms. It is obtained primarily by extracting natural gas from underground geological deposits, where it is the main component. It is also produced through the decomposition of organic matter by bacteria in environments without oxygen, such as wetlands, landfills, and livestock digestive systems. It is the simplest alkane and the main part of natural gas, used worldwide as a fuel. It is also a powerful greenhouse gas, with a present mixing ratio of 1.8 parts per million by volume and a global warming potential 25 times greater than carbon dioxide. Methane has a much shorter atmospheric lifetime (12 years) compared with carbon dioxide (5–200 years); so, it may be possible to regulate methane atmospheric concentrations through a reduction in emissions.

Regardless, methane is a very important source of energy which greatly contributes to the socio-economic activities of the world, therefore, making its extraction very paramount. The focus is on how to harness it into a useful substance and to curtail its emission into the atmosphere.

The oil and gas industries are dealers of methane and are major contributors to the extraction, processing, trading and emitting of methane. Their activities should be regulated and monitored for proper mitigation and abatement of the methane emission and exacerbating its economic benefit.

1.2 PROBLEM STATEMENT

Methane emissions represent a significant challenge in global efforts to mitigate climate change due to methane's strong heat-trapping ability and its contribution to near-term global warming. Emissions arise from several sources, including oil and gas production, processing and transportation, coal-related activities, and bioenergy, and may occur through operational processes such as venting, fugitive releases, and flaring-related methane emissions.

However, the magnitude and composition of methane emissions are not uniform across countries, regions, or industrial subsectors. This variation makes it difficult to determine where emissions are most concentrated, which activities contribute most significantly, and which emission pathways should receive priority for mitigation. Without systematically analyzing these differences, mitigation efforts may fail to target the countries, sectors, and emission sources where interventions could have the greatest impact.

Therefore, this study analyses global methane emissions data for 2022 to identify major emitting countries and regions, examine the contribution of different subsectors and emission reasons, and highlight significant emission patterns. Through the development of an interactive data dashboard, the analysis seeks to transform emissions data into actionable insights that can support evidence-based policy formulation, operational decision-making, emissions monitoring, and targeted methane mitigation strategies.

1.3 Aim of the Study

This study aims to analyze global methane emissions from the energy sector for the year 2022 by examining emission patterns across regions, countries, industrial subsectors, and emission reasons. The analysis seeks to identify major emission contributors and reveal the distribution of methane emissions across different geographical and operational categories. It is also aimed at identifying its climate impact using its CO_2 equivalent of different time horizons of global warming potential of 20-years and 100-years. The findings are intended to support evidence-based policy formulation, strategic planning, regulatory oversight,

emissions monitoring, and the development of effective methane mitigation strategies for the countries and regions represented in the dataset.

1.4 Objectives of the Study

The specific objectives of this study are to:

- Analyze global methane emissions from the energy sector for the year 2022.
- Identify the regions and countries contributing the highest and lowest methane emissions.
- Examine the distribution of methane emissions across different industrial subsectors within the energy sector.
- Assess the contribution of the various methane emission reasons, including flared, fugitive, and vented emissions, to overall methane emissions.
- Compare methane emission patterns across countries, regions, subsectors, and emission reasons to identify significant trends and emission hotspots.
- Calculate the CO_2 equivalence of the emission impact under GWP20 and GWP100 using the intergovernmental panel on climate change factor of 81.2 and 27.9 respectively.
- Develop an interactive Microsoft Excel dashboard to effectively visualize methane emission data and facilitate data-driven analysis.
- Provide evidence-based insights and recommendations to support methane emission monitoring, environmental policy formulation, regulatory decision-making, and methane mitigation strategies.

1.5 Scope of the Study

This study focuses on the analysis of global methane emissions from the energy sector and its corresponding climate impact using emissions data recorded for the 2022 reporting year. The analysis examines methane emission patterns across countries, regions, industrial subsectors, and emission reasons to identify major emission contributors and understand the distribution of methane emissions within the global energy landscape.

The study is limited to the variables contained in the dataset, including Country, Region, Emissions (Kt), Subsector, Emission Reason, and Year. The assessment covers key emission reasons such as flared, fugitive, and vented emissions, as well as methane emissions aggregated under the "All" category where applicable.

To facilitate data exploration and interpretation, an interactive dashboard was developed using Microsoft Excel, incorporating PivotTables, PivotCharts, slicers, dynamic charts, and Key Performance Indicators (KPIs). The dashboard provides

an interactive platform for visualizing methane emission patterns and comparing emission contributions across multiple dimensions.

The scope of this study is restricted to descriptive and comparative analysis based on the available dataset and does not include predictive modelling, forecasting, or the evaluation of methane emission trends across multiple years.

1.6 Significance of Study

Methane is one of the most potent greenhouse gases contributing to global warming, making the identification and management of methane emission sources a critical component of climate change mitigation efforts. By providing a comprehensive analysis of methane emissions across countries, regions, industrial subsectors, and emission reasons, this study contributes to a better understanding of the factors driving methane emissions within the global energy sector.

The findings of this study provide valuable insights for government agencies, regulatory authorities, policymakers, oil and gas companies, environmental organizations, researchers, and investors by identifying emission hotspots and highlighting areas where mitigation efforts can achieve the greatest impact. Such information can support the development of targeted policies, improved regulatory frameworks, and strategic investment decisions aimed at reducing methane emissions.

Furthermore, the interactive dashboard developed as part of this study demonstrates how business intelligence and data visualization tools can transform complex environmental datasets into accessible and actionable information. This enhances evidence-based decision-making by enabling stakeholders to monitor methane emissions, compare performance across regions and subsectors, and identify priority areas for operational improvements and environmental intervention.

Finally, this study illustrates the practical application of data analytics in addressing real-world energy and environmental challenges, highlighting the role of data-driven insights in supporting sustainable energy development and global efforts to reduce greenhouse gas emissions.

CHAPTER TWO

2.0 TECHNICAL BACKGROUND

2.1 Methane and the Energy Sector

According to International Energy Agency (IEA), 60% percent of methane emission is anthropogenic; Agriculture being the highest source of emission followed by the energy sector. The energy sector is therefore particularly relevant to methane mitigation because methane can be released at multiple stages of the fossil-fuel value chain. In oil and gas operations, emissions can occur during exploration and production, processing, storage, transmission and distribution. These emissions may arise intentionally through practices such as venting, unintentionally through equipment and infrastructure leaks, or through incomplete combustion associated with flaring.

Methane management has become an increasingly important component of energy and climate policy. UNEP identifies the fossil-fuel sector as having significant opportunities for technically feasible methane reductions, including measures targeting oil and gas operations.

2.2 Sources of Methane Emissions

Methane emissions can broadly be classified according to their source and the activity responsible for their release. At the global level, major anthropogenic sources include agriculture, fossil fuels, and waste. UNEP's Global Methane Assessment estimates that agriculture accounts for approximately 40% of anthropogenic methane emissions, fossil fuels approximately 35%, and waste approximately 20%. Within the fossil-fuel category, oil and gas operations and coal mining constitute important sources.

For this study, the analysis focuses on energy-sector subsectors represented in the dataset. These include:

- Onshore oil
- Onshore gas
- Offshore oil
- Offshore gas
- Gas pipelines and LNG facilities
- Steam coal
- Coking coal
- Bioenergy

- Other oil and gas sources
- Other coal sources
- Satellite-detected large oil and gas emissions

The inclusion of these categories allows methane emissions to be examined beyond a simple country-level comparison. It permits identification of the specific energy activities and subsectors responsible for the recorded emissions.

This distinction is important because a country with high total methane emissions may not necessarily have the same dominant emission source as another country. Consequently, effective methane-abatement strategies require consideration of where emissions occur within the energy value chain, rather than relying exclusively on national emission totals.

2.3 Emission Reasons: Flared, Fugitive, and Vented Methane

Beyond the industrial subsector in which methane is released, the operational pathway through which the release occurs is equally important for designing effective mitigation measures. The dataset used in this study classifies energy-sector methane releases into three principal emission reasons, in addition to an aggregated "All" category applied where a specific reason could not be isolated in the source data.

- Vented emissions – methane released directly and intentionally into the atmosphere as part of normal operational design, such as during equipment blowdowns, pneumatic device operation, or storage tank venting. Because the gas is released unburned, vented methane carries the full climate impact of methane itself, making it the most consequential emission pathway from a warming-potential standpoint.
- Fugitive emissions – unintentional leaks arising from imperfect equipment seals, valves, flanges, connectors, and ageing infrastructure across production, processing, and transmission systems. Fugitive emissions are typically diffuse and harder to detect than vented or flared releases, which makes leak detection and repair (LDAR) programmes central to their control.

- Flared emissions – methane released during the deliberate combustion of associated or unwanted gas at production sites, typically where there is insufficient infrastructure to capture, process, or transport the gas. Flaring converts most of the methane to CO2 and water vapour, but incomplete combustion allows a residual fraction of unburned methane to escape.

Records classified under the aggregated "All" category represent emissions that could not be attributed to a single reason in the source data (for example, several Bioenergy and coal-related entries). These records are retained in the overall emissions totals but are excluded from the reason-specific breakdown presented in Section 4.5, consistent with standard practice for handling unattributed categories in emissions accounting.

2.4 Global Warming Potential and the Climate Impact of Methane (GWP20 and GWP100)

Because different greenhouse gases trap heat with different efficiency and persist in the atmosphere for different lengths of time, a common metric is required to compare their climate impact on a consistent basis. The Global Warming Potential (GWP) expresses the cumulative radiative forcing of a given mass of a gas relative to the same mass of carbon dioxide (CO2), evaluated over a specified time horizon, with the result expressed as CO2-equivalent (CO2e).

This study applies the Intergovernmental Panel on Climate Change Sixth Assessment Report (IPCC AR6) GWP factors for methane: 81.2 over a 20-year horizon (GWP20) and 27.9 over a 100-year horizon (GWP100). The two horizons serve different analytical purposes. GWP100 is the horizon conventionally used in national greenhouse gas inventories and long-term climate targets, while GWP20 better reflects methane's outsized contribution to near-term warming, owing to its relatively short atmospheric lifetime of approximately 12 years compared with CO2. Because the same physical mass of methane is converted using a larger factor under the 20-year horizon, GWP20-based CO2e values are consistently larger than GWP100-based values for an identical emission — in this dataset, by a constant ratio of approximately 2.91 (81.2 ÷ 27.9), reflecting the fact that both figures describe the same underlying methane mass viewed over different time windows, rather than an additional physical quantity of gas.

Reporting both horizons allows the analysis to serve two audiences: policymakers concerned with long-term, cumulative climate commitments (best served by

GWP100), and those concerned with near-term climate tipping points and the urgency of methane abatement within the current decade (best served by GWP20).

CHAPTER THREE

3.0 METHODOLOGY

3.1 Research Design

This study adopts a descriptive and comparative research design based on secondary data. Descriptive analysis is used to summarise the volume and distribution of 2022 methane emissions across regions, countries, subsectors, and emission reasons, while comparative analysis is used to examine how emission patterns differ between countries, subsectors, and reporting categories. The study does not employ inferential statistics or predictive modelling; its objective is to characterise the existing emissions landscape and present it through an interactive dashboard rather than to forecast future emissions.

3.2 Data Source and Description

The dataset used in this study was obtained from a publicly available 2022 global methane emissions dataset sourced via Kaggle, compiled from energy-sector emissions monitoring and reporting covering the oil, gas, coal, and bioenergy value chains. The dataset comprises 1,119 records spanning 98 countries across 8 global regions (Africa, Asia Pacific, Central and South America, Europe, Middle East, North America, Russia and Caspian, and a residual "Other" grouping), disaggregated across 11 industrial subsectors and classified by emission reason.

3.3 Data Variables and Structure

Each record in the dataset is structured around the following variables:

- Region – the global region to which the reporting country belongs.
- Country – the reporting country.
- Emissions (Kt) – the mass of methane emitted, in kilotonnes.
- Sub Sector – the energy-sector activity responsible for the emission (e.g., Onshore Oil, Steam Coal, Gas Pipelines and LNG Facilities).
- Reason – the operational pathway of release (Flared, Fugitive, Vented, or All).

- CO2e GWP100 (IPCC AR6) and CO2e GWP20 (IPCC AR6) – the CO2-equivalent climate impact of each emission record under the 100-year and 20-year global warming potential horizons, respectively, together with each record's percentage contribution to the corresponding global total.

3.4 Tools and Software

All data cleaning, analysis, and visualisation for this study were carried out in Microsoft Excel. The workbook makes use of PivotTables and PivotCharts to summarise and cross-tabulate emissions by region, country, subsector, and emission reason; Slicers to enable interactive filtering by country and region; and Key Performance Indicator (KPI) cards to surface headline figures — total 2022 methane emissions, number of reporting regions and countries, and the highest-emitting region, country, and subsector — at a glance.

3.5 Dashboard Design and Analytical Approach

The analytical workflow proceeded in four stages. First, the raw dataset was reviewed and organised by region, country, subsector, and emission reason. Second, PivotTables were constructed to aggregate emissions along each of these dimensions, including a cross-tabulation of country against emission reason to support country-level diagnostic analysis. Third, PivotCharts were built directly from these PivotTables — including a Top-10-countries ranking, a regional and subsector breakdown, and a Flared/Fugitive/Vented composition chart — so that the visuals remain linked to, and refreshable from, the underlying data. Fourth, these elements were consolidated into a single-page interactive dashboard combining KPI summary cards, comparative bar charts, a composition pie chart, and a country-selection slicer, allowing a reader to move from a high-level global summary down to an individual country's emission profile without leaving the dashboard view.

3.6 Limitations of the Methodology

As a descriptive, cross-sectional analysis of a single reporting year (2022), this study does not capture year-on-year trends in methane emissions and cannot, on its own, indicate whether emissions from a given country or subsector are rising,

falling, or stable. The analysis also relies on secondary, previously reported and/or estimated emissions data; reporting completeness and estimation methods (including satellite-detected emissions) are not uniform across all 98 countries, and any inconsistencies in the source data are inherited by this analysis. Finally, the dashboard's interactivity (slicers and PivotChart filtering) requires the workbook to be viewed in Microsoft Excel; a static export (PDF or image) does not preserve this interactivity.

CHAPTER FOUR

4.0 RESULTS AND DISCUSSION OF FINDINGS

4.1 Overview of 2022 Global Methane Emissions

The dataset records a total of 135,347.17 kilotonnes (kt) of methane emitted from the global energy sector in 2022, spanning 98 countries across 8 regions and 11 industrial subsectors. This total forms the denominator against which all regional, country-level, subsector, and emission-reason shares discussed in this chapter are calculated, and corresponds to the headline KPI figure displayed on the dashboard.

4.2 Regional Distribution of Methane Emissions

Table 4.1 presents total 2022 methane emissions by region, ranked from highest to lowest.

Region	Emissions (Kt)	Share of Global Total
Asia Pacific	41,620.09	30.75%
Russia & Caspian	26,662.86	19.70%
North America	20,899.36	15.44%
Middle East	16,270.41	12.02%
Africa	13,267.94	9.80%
Central and South America	7,323.16	5.41%
Europe	6,882.48	5.09%
Other	2,420.88	1.79%

Table 4.1: 2022 Methane Emissions by Region

Asia Pacific is the highest-emitting region, accounting for approximately 30.75% of global methane emissions recorded in the dataset, driven substantially by China's oil, gas, and coal operations (Section 4.3). Russia and Caspian follows as the second-highest region, reflecting the scale of Russia's and Turkmenistan's gas

pipeline and onshore gas infrastructure. Together, the top three regions — Asia Pacific, Russia & Caspian, and North America — account for close to two-thirds (65.9%) of total recorded emissions, indicating that global methane mitigation efforts targeting these three regions alone could address the majority of energy-sector methane emissions captured in this dataset.

4.3 Country-Level Distribution of Methane Emissions

Table 4.2 presents the ten highest-emitting countries in the dataset.

Rank	Country	Emissions (Kt)
1	China	25,372.22
2	Russia	18,078.08
3	United States	16,769.63
4	Iran	5,629.82
5	India	5,185.24
6	Indonesia	5,080.37
7	Turkmenistan	4,740.04
8	Nigeria	3,306.07
9	Saudi Arabia	3,051.10
10	European Union	2,760.97

Table 4.2: Top 10 Methane-Emitting Countries, 2022

China is the single largest national emitter, contributing approximately 18.75% of global methane emissions in the dataset — roughly 40% more than Russia, the second-highest emitter. China, Russia, and the United States together account for approximately 44.6% of global methane emissions, underscoring the concentration of emissions among a small number of major energy-producing economies. Nigeria appears among the top ten emitters at 3,306.07 kt, the highest-ranked country from Africa, which is of particular relevance to national and regional methane monitoring and mitigation policy within the energy sector.

4.4 Sub-sector Contribution to Methane Emissions

Table 4.3 presents the distribution of 2022 methane emissions across the eleven industrial subsectors captured in the dataset.

Sub Sector	Emissions (Kt)	Share of Global Total
Onshore oil	34,332.78	25.37%
Steam coal	30,047.27	22.20%
Onshore gas	18,097.60	13.37%
Gas pipelines and LNG facilities	12,639.02	9.34%
Coking coal	10,313.16	7.62%
Bioenergy	9,926.41	7.33%
Offshore oil	8,349.61	6.17%
Offshore gas	4,342.40	3.21%
Satellite-detected large oil and gas emissions	3,106.40	2.30%
Other from oil and gas	2,493.18	1.84%
Other from coal	1,699.34	1.26%

Table 4.3: Methane Emissions by Industrial Sub-Sector, 2022

Onshore Oil is the single largest subsector, accounting for approximately a quarter (25.37%) of global methane emissions, followed closely by Steam Coal (22.20%) and Onshore Gas (13.37%). Together, these three subsectors account for approximately 60.9% of total recorded emissions. Notably, coal-related activities (Steam Coal and Coking Coal combined) contribute 29.82% of total emissions — a larger share than any single oil or gas subsector — indicating that methane mitigation strategies focused exclusively on oil and gas operations, without addressing coal-mine methane, would leave a substantial share of energy-sector emissions unaddressed.

4.5 Distribution by Emission Reason

Excluding records classified under the aggregated "All" category (Section 2.3), the dataset attributes methane emissions to three specific reasons, summarised in Table 4.4.

Emission Reason	Emissions (Kt)	Share (excl. "All")
Vented	52,356.90	67.33%
Fugitive	17,729.18	22.80%
Flared	7,675.33	9.87%

Table 4.4: Methane Emissions by Emission Reason, 2022 (excluding "All")

Vented emissions dominate the emission-reason breakdown, accounting for more than two-thirds (67.33%) of attributable methane emissions. Because vented methane is released unburned and by deliberate operational design, this finding indicates that the single highest-leverage mitigation pathway available to operators is the reduction or elimination of routine venting — for example, through vapour recovery units, pneumatic device replacement, and tighter operational standards — rather than flaring reduction or leak repair alone, although both remain important secondary measures given their combined 32.67% share.

4.6 Country-Level Deep Dive by Emission Reason

To illustrate how emission-reason composition varies even among the highest-emitting countries, Table 4.5 presents the Flared, Fugitive, and Vented breakdown for the five largest national emitters.

Country	Flared (Kt)	Fugitive (Kt)	Vented (Kt)
United States	458.3	3,885.4	9,241.2
Russia	1,200.2	2,885.3	8,368.9
Iran	1,260.2	1,031.4	3,169.1
China	68.4	906.3	2,007.2
India	72.5	278.0	597.1

Table 4.5: Emission Reason Breakdown for the Top Five Emitting Countries, 2022

Vented emissions dominate the profile of every one of the five largest emitters, consistent with the global pattern observed in Section 4.5. However, the relative importance of flaring differs sharply between countries: Iran and Russia show substantially higher flared volumes than China or India, indicating comparatively greater reliance on flaring as a disposal method for associated gas in the former group, while China's profile is overwhelmingly dominated by fugitive and vented releases rather than flaring. This variation demonstrates that a single, uniform mitigation strategy applied across all countries would be less effective than country-specific interventions informed by each nation's dominant emission pathway.

4.7 Climate Impact: CO2-Equivalent Emissions under GWP20 and GWP100

Applying the IPCC AR6 conversion factors described in Section 2.4, the 135,347.17 kt of methane emitted in 2022 is equivalent to approximately 3,776,186.10 kt of CO2 (3.78 million kt CO2e) under the 100-year horizon (GWP100), and approximately 10,990,190.36 kt of CO2 (10.99 million kt CO2e) under the 20-year horizon (GWP20). The consistent ratio of 2.91 between the two totals reflects the fixed ratio of the underlying GWP factors (81.2 ÷ 27.9) rather than any difference in the physical quantity of methane emitted. This near three-fold difference illustrates the scale of methane's near-term warming contribution relative to its long-term, CO2-equivalent accounting, and reinforces why climate frameworks increasingly track methane using both time horizons rather than GWP100 alone.

4.8 Dashboard Design and Interactive Features

The findings summarised in Sections 4.1 to 4.7 are presented through a single interactive Microsoft Excel dashboard, comprising: (i) KPI summary cards displaying total 2022 methane emissions, the number of reporting regions and countries, and the highest-emitting region, country, and subsector; (ii) a Top-10-countries PivotChart; (iii) a regional and subsector emissions breakdown; (iv) a Flared/Fugitive/Vented composition pie chart with an accompanying methodological note on the treatment of the "All" category; (v) a GWP20-versus-GWP100 climate-impact comparison chart; and (vi) a country-selection slicer enabling a reader to filter the subsector and emission-reason charts down to an individual country of interest. This design allows the dashboard to serve both as a

high-level summary tool for policy and management audiences and as a drill-down diagnostic tool for more technical, country-specific analysis.

4.9 Summary of Key Findings

- Global 2022 methane emissions from the energy sector totalled 135,347.17 kt across 98 countries and 8 regions.
- Asia Pacific is the highest-emitting region (30.75%), and China is the highest-emitting country (18.75% of the global total).
- Onshore Oil, Steam Coal, and Onshore Gas together account for approximately 60.9% of global emissions; coal-related subsectors combined (29.82%) exceed any single oil or gas subsector.
- Vented emissions dominate the emission-reason profile globally (67.33%) and across every one of the top five emitting countries.
- Flaring's relative importance varies significantly by country, with Iran and Russia showing markedly higher flared shares than China or India.
- 2022 methane emissions are equivalent to approximately 3.78 million kt of CO2 under GWP100, or 10.99 million kt of CO2 under GWP20 — a 2.91-fold difference reflecting methane's outsized near-term climate impact.

CHAPTER FIVE

5.0 SUMMARY, CONCLUSION AND RECOMMENDATIONS

5.1 Summary

This study set out to analyse global methane emissions from the energy sector for the year 2022, examining how emissions are distributed across regions, countries, industrial subsectors, and emission reasons, and to express their climate impact in CO2-equivalent terms under both 20-year and 100-year global warming potential horizons. Using a dataset of 1,119 records covering 98 countries in 8 regions, the analysis found that global 2022 methane emissions totalled 135,347.17 kt, dominated by the Asia Pacific region and China at the country level, by Onshore Oil and coal-related activities at the subsector level, and by vented releases at the emission-reason level. These findings were consolidated into an interactive Microsoft Excel dashboard combining KPI cards, PivotCharts, and a country-selection slicer, fulfilling the study's aim of transforming raw emissions data into an accessible, decision-support tool.

5.2 Conclusion

The analysis demonstrates that global energy-sector methane emissions are highly concentrated, both geographically and operationally. A small number of countries — China, Russia, and the United States — account for nearly half of all recorded emissions, and a small number of subsectors and emission pathways, particularly Onshore Oil, coal mining, and vented releases, account for a disproportionate share of the total. This concentration is, in itself, an encouraging finding for policy design: it indicates that targeted interventions aimed at a limited set of countries, subsectors, and operational practices — rather than uniform, undifferentiated global regulation — have the potential to deliver a substantial reduction in global methane emissions. The study also confirms the importance of reporting methane's climate impact under both GWP20 and GWP100, as reliance on the 100-year horizon alone understates methane's contribution to near-term warming by a factor of approximately 2.91.

5.3 Recommendations

Based on the findings of this study, the following recommendations are proposed:

- Prioritise venting reduction as the primary methane mitigation lever, given that vented releases account for over two-thirds of attributable emissions globally and across every major emitting country; this includes accelerating adoption of vapour recovery units and low- or zero-emission pneumatic devices.
- Target mitigation policy and investment toward the highest-emitting countries and subsectors identified in this study — particularly Onshore Oil operations in China, Russia, and the United States, and coal-mine methane management in major coal-producing economies — to achieve the greatest emissions reduction per unit of regulatory or investment effort.
- Extend leak detection and repair (LDAR) programmes and infrastructure modernisation in countries and subsectors where fugitive emissions form a significant share of the total, to reduce unintentional losses across gas pipelines, LNG facilities, and onshore/offshore processing infrastructure.
- Where flaring remains necessary, invest in flare-gas capture and combustion-efficiency improvements, particularly in countries such as Iran and Russia where flared emissions represent a comparatively larger share of the national emissions profile.
- Adopt satellite-based and other remote-sensing monitoring, reporting, and verification (MRV) systems to improve the consistency and completeness of country-level methane reporting, addressing the data-quality limitations noted in Section 3.6.
- Report methane's climate impact using both GWP20 and GWP100 in national and corporate disclosures, to ensure that near-term climate urgency is not obscured by the longer-term framing of GWP100 alone.

5.4 Limitations of the Study

This study is subject to several limitations. First, it is a cross-sectional analysis of a single reporting year (2022) and does not capture emission trends over time; a country or subsector's 2022 ranking should not be read as indicative of a persistent or worsening trajectory without multi-year data. Second, the analysis is restricted to the energy sector and does not include agriculture and waste, which together account for a larger share of total anthropogenic methane emissions globally (Section 2.2); the findings should therefore be interpreted as an energy-sector analysis rather than a complete national or global methane inventory. Third, the

study relies on secondary, previously reported and/or estimated data, including satellite-detected emissions, and inherits any inconsistencies in reporting standards, estimation methods, or completeness across the 98 countries covered. Finally, the interactivity of the dashboard (PivotChart filtering and slicers) is only available within Microsoft Excel; static exports of the dashboard do not retain this functionality.

5.5 Suggestions for Further Study

- Extend the analysis to a multi-year panel dataset to identify genuine emission trends, rather than a single-year snapshot, and to evaluate the effectiveness of mitigation measures over time.
- Incorporate agricultural and waste-sector methane emissions alongside the energy sector to produce a complete national and global methane inventory.
- Apply predictive or scenario-based modelling to estimate the emissions-reduction impact of specific interventions (e.g., LDAR programmes, flare-gas capture) identified in Section 5.3.
- Integrate the cost-effectiveness (marginal abatement cost) of different mitigation measures to support prioritisation of investment alongside the emissions-volume prioritisation presented in this study.

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